

Making Digital Resources with Quarto

CELPiE Summer Symposium - 18 June 2026

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Posit

From the company that brought you RStudio, R Markdown, and more recently Positron, comes



an open-source scientific and technical publishing system.

Why?

1. **Accessibility:** Creating digital resources solves a well known problem of screen readability of PDF documents. Solve this problem by publishing *simultaneously*
 - multiple formats: Word, PDF, HTML, etc.
2. **Flexibility:** write in markdown, publish as website, document, presentations, blog, dashboard, etc.
 - more authentic and creative modes of assessment (in the age of AI)
 - more modern, engaging material
3. **Reproducibility:** Incorporate reproducible code directly into your documents.

- along with data, tables, and figures
 - using a range of languages: R, Python, Julia, even software like Stata
 - sync changes through an online host (e.g. GitHub Pages)
4. **Interactivity:** Create “live” interactive documents that allow you to interact with data in real time.

These are a few of my favourite things...

#1 LaTeX without *LaTeX*

LaTeX is *beautiful*, but LaTeX can also be a huge waste of time. And it’s not really a skill our students need.

Binary Choice Models

1 Overview

In this handout we will review the binary case of discrete choice models (a type of limited dependent variable model). We will review:

- linear probability model (LPM);
- probit model;
- and logit model.

Studying the simple case binary choice models, will provide you with tools needed to explore richer models like multinomial logit/probit, ordered logit/probit, and conditional logit.

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- 2 Binary Outcomes
- 3 Linear Probability Model
- 4 Latent Variable Model
- 5 Index Model Approach
- 6 Estimation
- 7 Interpretation & Fit
- 8 References

Other Formats

PDF

Figure 1: Example from [EC910 lecture notes](#)

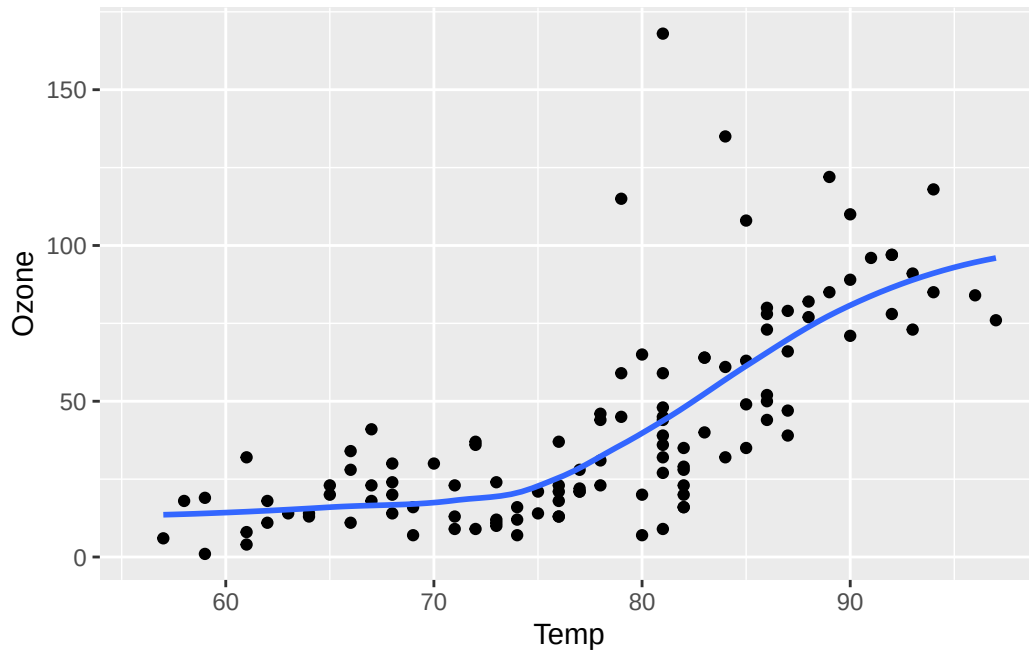
A nice feature of Quarto is that it supports LaTeX math syntax and will publish your documents PDFs simultaneously as HTML or Word, so you can have the best of both worlds.

#2 Incorporation of code

There is no better way to incorporate code into your teaching material, both from a design and feature perspective.¹

¹See this [example](#) from the [Quarto Gallery](#).

```
library(ggplot2)
ggplot(airquality, aes(Temp, Ozone)) +
  geom_point() +
  geom_smooth(method = "loess", se = FALSE)
```



#3 Multi-lingual documents

There are some [caveats](#), but you can write documents in multiple languages.

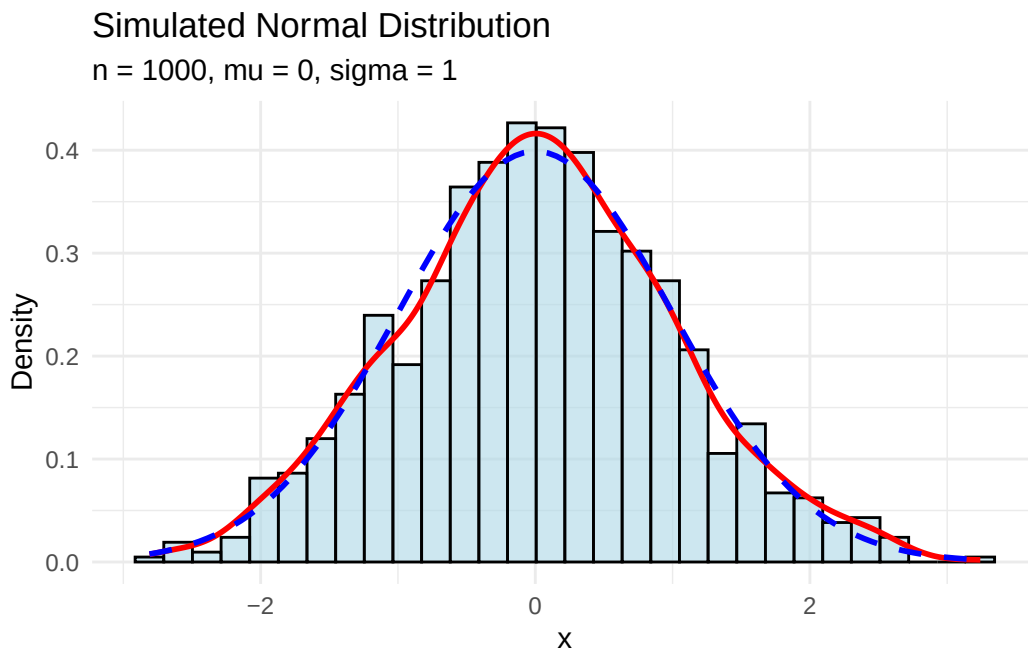
R

```
library(tibble)

set.seed(123)
n <- 1000
mu <- 0
sigma <- 1
df <- tibble(x = rnorm(n, mean = mu, sd = sigma))

gg <- ggplot(df, aes(x = x)) +
  geom_histogram(aes(y = ..density..), bins = 30, fill = "lightblue", color = "black", alpha
```

```
geom_density(color = "red", size = 1) +
stat_function(fun = dnorm, args = list(mean = mu, sd = sigma), color = "blue", linetype = "dashed") +
labs(
  title = "Simulated Normal Distribution",
  subtitle = paste0("n = ", n, ", mu = ", mu, ", sigma = ", sigma),
  x = "x",
  y = "Density"
) +
theme_minimal()
gg
```



Python

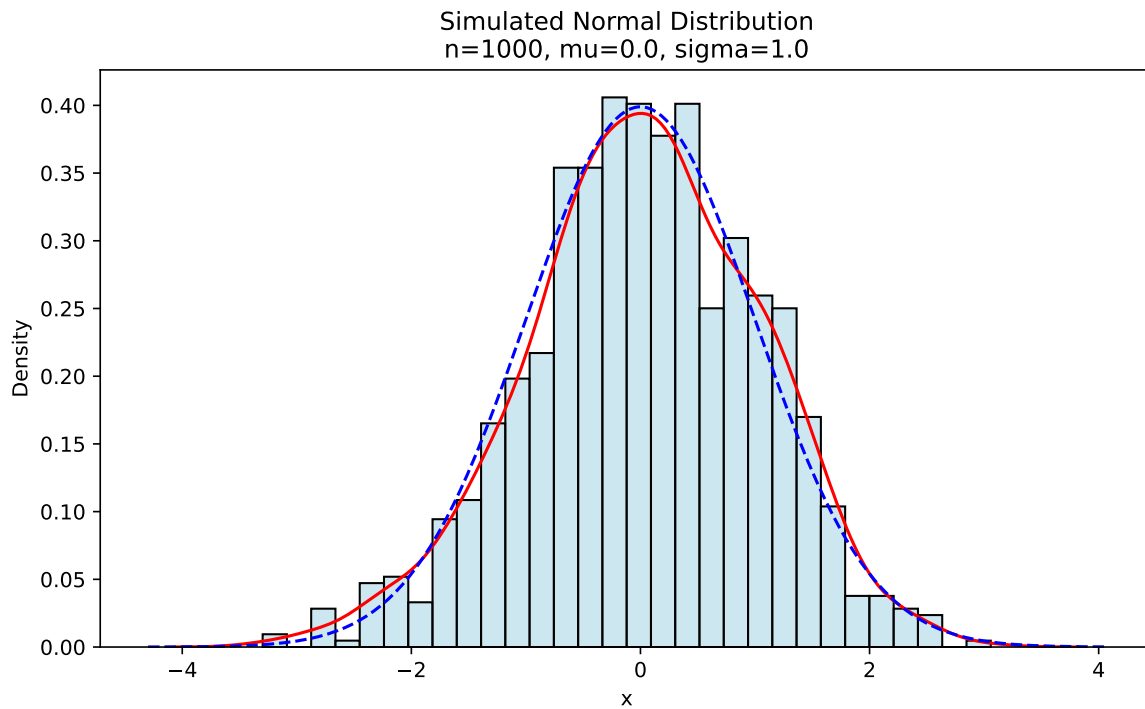
```
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

def plot_normal_sim(n=1000, mu=0.0, sigma=1.0, bins=30, seed=123):
    rng = np.random.default_rng(seed)
    x = rng.normal(loc=mu, scale=sigma, size=n)
    fig, ax = plt.subplots(figsize=(8, 5))
    sns.histplot(x, bins=bins, stat="density", color="lightblue", edgecolor="black", alpha=0.5)
```

```

sns.kdeplot(x, color="red", linewidth=1.5, ax=ax)
xs = np.linspace(x.min() - 1, x.max() + 1, 400)
pdf = (1.0 / (sigma * np.sqrt(2 * np.pi))) * np.exp(-0.5 * ((xs - mu) / sigma) ** 2)
ax.plot(xs, pdf, color="blue", linestyle="--", linewidth=1.5)
ax.set_title(f"Simulated Normal Distribution\nn={n}, mu={mu}, sigma={sigma}")
ax.set_xlabel("x")
ax.set_ylabel("Density")
fig.tight_layout()
return fig
plot_normal_sim()

```

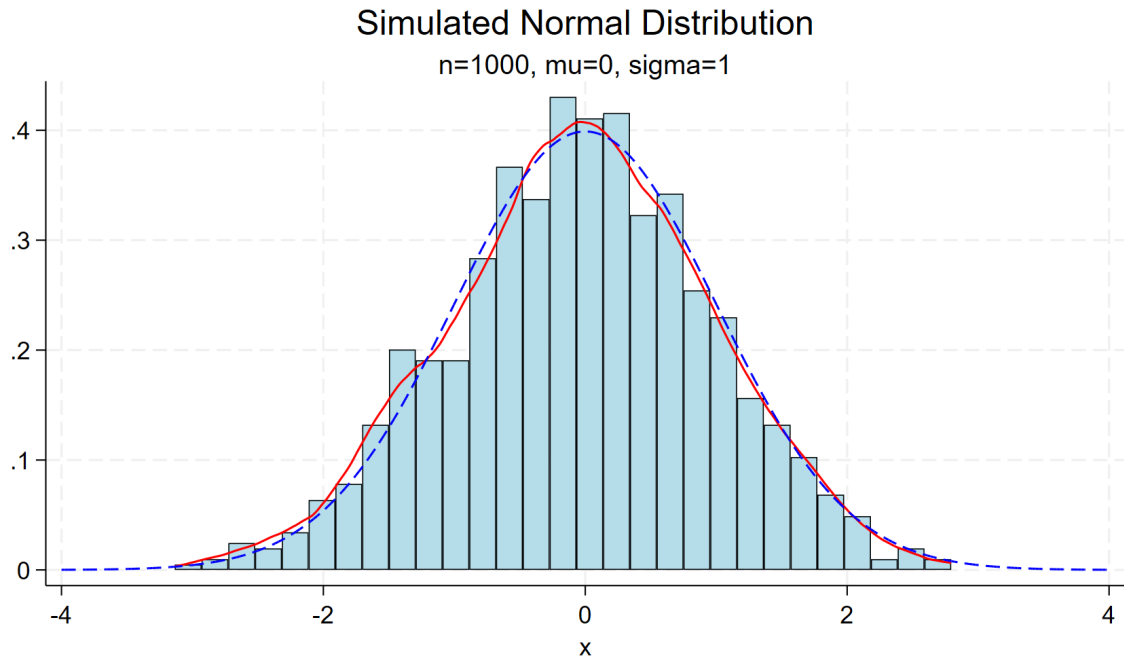


Stata

```

set obs 1000
gen x = rnormal()
twoway (histogram x, fcolor(ltblue) lcolor(black)) (kdensity x, lcolor(red)) (function y=normpdf(x,0,1))
graph export "img/pres-symp-jun26/stata_histogram.png", replace

```



#4 Slides vs lecture notes, have *both*!

Unsure about making slides or lecture notes, make both **simultaneously**.

.qmd

```
263- ## Risky workers are (almost always) a better bet
264
265 1. The relative value of the risky worker falls as  $r \uparrow$  (i.e. discounting future earnings more), but is always positive.
266
267 ::: {.callout-tip title="Example"}
268
269 -  $r=0.4$ 
270 -  $\theta=0.2$ 
271
272 :::
273
274 ```{r}
275 #| echo: false
276 #| code-fold: false
277
278 # Parameters
279 r <- 0.4
280
281 # Generic grid
282 df <- data.frame(
283   t = 1:T,
284   Y = Y,
285   x = rep(NA, T)
286 )
287
288 # Safe grid, with NPV calculation per period
289 dfs <- df
290 dfs$Q <- Qs
291 dfs$x <- (((1-theta)/(1+r))^(dfs$t-1))*(dfs$Q-dfs$Y)
292
293 dfr$x[1:tp] <- (((1 - theta)/(1 + r))^(dfr$t[1:tp] - 1))*rho*(dfr$Qp[1:tp] - dfr$Y[1:tp]) + (1 - rho)*(dfr$Ql[1:tp] - dfr$Y[1:tp])
294 dfr$x[(tp+1):T] <- (((1 - theta)/(1 + r))^(dfr$t[(tp+1):T] - 1))*rho*(dfr$Qp[(tp+1):T] - dfr$Y[(tp+1):T])
295
296 # Sum over all periods
297 NPVs <- sum(dfs$x)
298 NPVr <- sum(dfr$x)
299 cat("NPV of safe = ",NPVs,"and NPV of risky = ",NPVr,". Relative value of risky = ", NPVr/NPVs)
300
301
302 2. The relative value of the risky worker falls as  $\theta \uparrow$  (i.e. higher turnover), but is always positive.
303
```

HTML slides

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Risky workers are (almost always) a better bet

1. The relative value of the risky worker **falls** as $r \uparrow$ (i.e. discounting future earnings more), **but is always positive**.

Example

- $r = 0.4$
- $\theta = 0.2$

NPV of safe = 116.2336 and NPV of risky = 172.7268 . Relative value of risky = 1.486032

2. The relative value of the risky worker **falls** as $\theta \uparrow$ (i.e. higher turnover), **but is always positive**.

Example

- $r = 0.2$
- $\theta = 0.5$

NPV of safe = 85.70077 and NPV of risky = 108.0019 . Relative value of risky = 1.260221


```
#print(dfr)

# Sum over all periods
NPVr <- sum(dfr$x)
cat("NPV of risky = ",NPVr,". Relative value of risky = ", NPVr/NPVs)
```

NPV of risky = 243.4969 . Relative value of risky = 1.65196

Productivity

Note that risky and safe workers have the same *expected* profitability,

$$(Q_s - Y) = \rho(Q_r^p - Y) + (1 - \rho)(Q_r^l - Y)$$

💡 Example

$$(\pounds 100,000 - \pounds 50,000) = 0.5 \times (\pounds 300,000 - \pounds 50,000) + 0.5 \times (-\pounds 100,000 - \pounds 50,000)$$

Risky workers are (almost always) a better bet

1. The relative value of the risky worker falls as $r \uparrow$ (i.e. discounting future earnings more), but is always positive.

💡 Example

- $r = 0.4$
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NPV of safe = 116.2336 and NPV of risky = 172.7268 . Relative value of risky = 1.486032

2. The relative value of the risky worker falls as $\theta \uparrow$ (i.e. higher turnover), but is always positive.

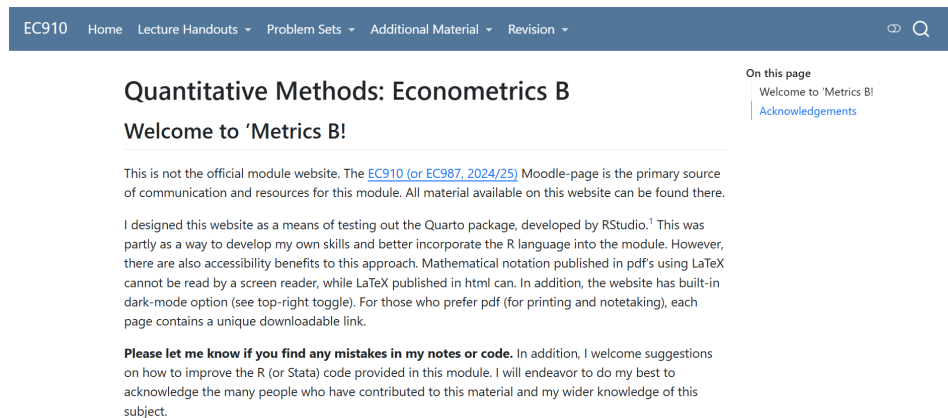
💡 Example

- $r = 0.2$
- $\theta = 0.5$

NPV of safe = 85.70077 and NPV of risky = 108.0019 . Relative value of risky = 1.260221

#5 Organization

Become a ‘web developer’ overnight and organize your materials as a website or e-book.



This has *huge* benefits for students. You can add links in Moodle directly to your website, which then remain up to date.²

Downsides

Here are a few things you will need to consider:

- Parsing settings to LaTeX can appear to be tricky! (I’ve not even tried, yet.)
- Formatting html documents requires the use of a CSS/SCSS style sheet.
 - AI can help with this.
- Markdown is an intentionally simple (easy-to-learn) language, but it is not as featureful as LaTeX.
 - Again, formatting comes through CSS.
 - My least favourite feature is footnotes!
- Coding with a markdown file requires the use of code blocks.
 - Not *necessarily* how you want your students to engage with code.
 - Can probably write a script to strip the code and create a .r/.py/.do script for your students to run.

²Quarto provides excellent guidance on how to host a site: <https://quarto.org/docs/websites/>

- For example, I created a script that turns multi-lingual html documents into separate language-specific PDFs: <https://neil-lloyd.github.io/digital-resources/>